

p -LESS SAMPLING: A ROBUST HYPERPARAMETER-FREE APPROACH FOR LLM DECODING

Runyan Tan^{1,2,3}, Shuang Wu¹, Phillip Howard¹

¹Thoughtworks

²National University of Singapore

³Corresponding Author

{runyan.tan, shuang.wu, phillip.howard}@thoughtworks.com
run@nus.edu.sg

ABSTRACT

Obtaining high-quality outputs from Large Language Models (LLMs) often depends upon the choice of a sampling-based decoding strategy to probabilistically choose the next token at each generation step. While a variety of such sampling methods have been proposed, their performance can be sensitive to the selection of hyperparameters which may require different settings depending upon the generation task and temperature configuration. In this work, we introduce p -less sampling: an information-theoretic approach to sampling which dynamically sets a truncation threshold at each decoding step based on the entire token probability distribution. Unlike existing methods, p -less sampling has no hyperparameters and consistently produces high-quality outputs as temperature increases. We provide theoretical perspectives on p -less sampling to ground our proposed method and conduct experiments to empirically validate its effectiveness across a range of math, logical reasoning, and creative writing tasks. Our results demonstrate how p -less sampling consistently outperforms existing sampling approaches while exhibiting much less degradation in text quality at higher temperature values. We further show how p -less achieves greater inference-time efficiency than alternative methods through lower average token sampling times and shorter generation lengths, without sacrificing accuracy. Finally, we provide analyses to highlight the benefits of p -less through qualitative examples, case studies, and diversity assessments. The code is available at [🔗](#).

1 INTRODUCTION

The increasingly impressive capabilities exhibited by Large Language Models (LLMs) in recent years have been aided by advancements in sampling-based decoding strategies which probabilistically select tokens at each generation step. In contrast to deterministic methods such as greedy decoding and beam search (Freitag & Al-Onaizan, 2017), sampling-based strategies can produce more diverse and human-like language outputs while avoiding issues such as neural text degeneration (Holtzman et al.). Truncation-based sampling strategies (e.g., top- p (Holtzman et al.), top- k (Fan et al., 2018b), ϵ -sampling (Freitag et al., 2023), mirostat (Basu et al., 2021), min- p (Nguyen et al., 2024)) have proven particularly effective by truncating the token probability distribution to only a subset of higher-likelihood tokens before sampling. However, the truncation thresholds produced by these existing approaches depend upon the specification of hyperparameters whose optimal values can vary depending upon the generation task and sampling temperature (Zhou et al., 2024).

To address this deficiency, we introduce p -less sampling: a parameter-less sampling strategy grounded in information theory. p -less sampling uniquely possesses several desirable properties for a sampling-based decoding method. Specifically, p -less sampling is a distribution-aware approach which dynamically adjusts the truncation threshold at each time step using the entire token probability distribution. In this way, p -less provides a more principled, information-theoretic approach to determining the truncation threshold than alternative methods. The truncation threshold produced by p -less also dynamically changes with temperature, producing robust results even at high temperature settings where

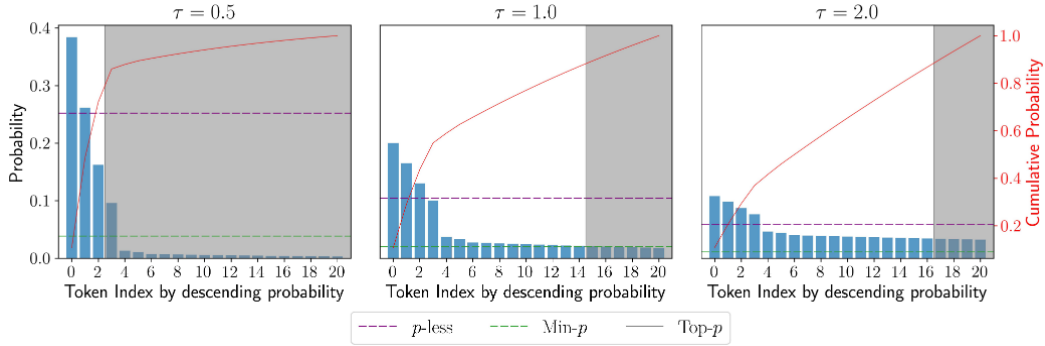


Figure 1: Comparison of truncation thresholds produced by p -less, $\text{min-}p$, and $\text{top-}p$ for a token probability distribution with different applied temperatures (τ). As temperature increases, p -less avoids admitting a large number of lower-likelihood tokens by considering the entropy of the distribution in computing the threshold.

other sampling approaches suffer from text degeneration. An illustration of this effect is provided in Figure 1, which shows how p -less sensibly truncates the long-tail of lower probability tokens even at higher temperatures which flatten the probability distribution. In contrast, other approaches which do not consider the entropy of the entire probability distribution admit far more tokens at high temperature values, thereby leading to greater degeneracy. We include examples of token distributions in Appendix B.6 to show the effects of temperature, vocabulary size and distribution profiles on the truncation threshold, including the ability of p -less to admit tokens from long-tailed distributions.

We provide theoretical perspectives on p -less sampling and ground our approach by interpreting it in connection to the family of Rényi entropies (Rényi, 1961). Through extensive experiments, we demonstrate the strong performance of p -less sampling across five math, logical reasoning, and creative writing datasets using three LLMs spanning multiple model sizes and architectures. Our results show that p -less sampling excels at math and reasoning tasks, consistently achieving high accuracy across a wide range of temperature values. In contrast, other evaluated sampling approaches have significantly greater variability in performance with respect to temperature, often exhibiting declines in accuracy at high temperature values. Our p -less approach also provides similar benefits for creative writing, producing the best performance in automated evaluations for the writing prompts dataset. We further validate these results through a human evaluation study, finding that human annotators also prefer stories generated by p -less over alternative sampling approaches. Finally, we demonstrate the superior inference-time efficiency provided by p -less sampling and conduct additional analyses on text diversity, qualitative examples, and case studies.

To summarize, our main contributions are as follows:

1. We introduce p -less sampling: a robust parameter-less sampling strategy grounded in information theory
2. Through extensive experiments, we validate the effectiveness of p -less sampling using three LLMs and five datasets spanning math, logical reasoning, and creative writing tasks.
3. We show that p -less is more efficient than other methods, both in terms of token sampling speed and overall generation length, without sacrificing task-specific performance.
4. We provide additional analyses to highlight the benefits of p -less through the lens of text diversity, qualitative examples, and case studies.

2 RELATED WORK

Text decoding in LLMs involves a truncation process where low probability tokens or those expected to disrupt coherence are excluded from the sampling pool. $\text{Top-}k$ (Fan et al., 2018a) restricts sampling to the k most probable tokens, which could lead to incoherent generations when the distribution is extremely uniform or peaked. $\text{Top-}p$ (Holtzman et al.) improves upon this by sampling from the smallest set of tokens whose cumulative probability exceeds a threshold p . ϵ -sampling (Freitag et al.,

2023) proposes to truncate all tokens with probabilities below a cut-off threshold ϵ quantity. Both top- p and ϵ -sampling remain lacking in adapting to high-entropy conditions, which is typical when temperature is turned up, such as for use cases where diversity is preferred. Similar to ϵ -sampling, η -sampling (Hewitt et al., 2022) proposes an entropy-aware variant which defines the threshold as the minimum of ϵ and a scaled negative Shannon entropy exponential quantity. This however introduces additional hyperparameters and relies on the assumption that the entropy follows a uniform distribution baseline.

A more sophisticated approach in the form of mirostat (Basu et al., 2021) assumes that the token distribution follows Zipf’s Law and dynamically adjusts the threshold to maintain a target surprisal. This dynamic feedback involves task and model dependent tuning of the target surprisal and a learning rate. The min- p method (Nguyen et al., 2024) attempts to simplify truncation through a fractional hyperparameter, which is multiplied by the modal probability to define the truncation threshold. While being more empirically robust at high temperatures, min- p remains sensitive to the choice of the fractional hyperparameter, and leverages only a single statistic from the token probability distribution. Another truncation approach aimed at balancing diversity with coherence (Zhu et al., 2024) increments the sampling set until the improvement in confidence falls short of an error term; it is similarly sensitive to the choice of the error hyperparameter and additionally requires iterative computations for differences and comparisons. In contrast, our proposed p -less sampling method avoids these pitfalls by operating directly on the empirical token distribution without imposing parametric assumptions or requiring hyperparameter tuning. This approach offers a model-agnostic threshold which is robust in high-entropy regimes.

Beyond truncation-based sampling techniques, a variety of decoding approaches have been proposed to improve the quality of text produced by LLMs. Contrastive decoding (Li et al., 2022) aims to improve text quality by contrasting token predictions from an expert and amateur model, with the intent of maximizing expert-like generations while minimizing amateur-like text. (Su et al., 2022) introduces contrastive search decoding, which augments top- k with an additional degeneration penalty hyperparameter α that balances the candidate token confidence with greater dissimilarity between the previous context and the context continuation using the token. (Arias et al., 2024) advances this strategy through examining local variations in the entropy to determine the top- k and degeneration penalty hyperparameters, thereby improving robustness across diverse contexts. While this strategy adaptively proposes these hyperparameters at each generation step, a temperature factor q is required and an implicit centering choice of the k quantity is made via the introduction of a pair of bounding hyperparameters. (Ding et al., 2025) further incorporates the global entropy dynamics to stabilize decoding under sudden local entropy spikes and guarantees an unbiased estimator for the instantaneous entropy, offering resilience against volatility and eliminating the need for the temperature factor in (Arias et al., 2024); however it introduces two hyperparameters, λ for computing the degeneration penalty and w to specify the window size for examining variations in local entropy. Other controlled decoding methods such as Neurologic Decoding (Lu et al., 2020) constrain text generation to achieve various objectives such as improved diversity, which is particularly useful for applications like synthetic data generation (Howard et al., 2022; 2024; Rosenman et al., 2024). Arithmetic sampling (Vilnis et al., 2023) uses parallel sampling to improve beam sampling based on an arithmetic code book defined implicitly by the model; Parashar et al. (2024) demonstrate how arithmetic sampling produces more diverse generations than ancestral sampling across reasoning and translation tasks. While these methods have some overlapping aims as our work, they differ substantially in their level of complexity and can be viewed as complementary approaches that can be used in conjunction with p -less or other truncation-based sampling methods.

3 METHODOLOGY

3.1 THE p -LESS SAMPLING METHOD

At every time step t , an autoregressive language model infers a distribution of the vocabulary tokens conditioned on the token sequence already generated from step 1 to step $t - 1$. Essentially, p -less considers information from the entire probability distribution after the application of temperature; it computes the likelihood of a correct random guess given the distribution, which serves as our principled probability threshold adapted to the token distribution at every decoding step. We admit tokens into the sampling distribution whose likelihoods are at least that of the correct random guess

likelihood. Furthermore, to determine which and how many tokens to sample from, the p -less threshold varies in a meaningfully opposite direction with entropy; as entropy increases, more tokens with lower probability are admitted for sampling.

Formally, let $v \in \mathcal{V}$ denote the set of tokens from a vocabulary $\mathcal{V}$. At each time step t , let $\mathcal{P}(\mathcal{S} = v)$ denote the probability that token v is sampled and $\mathcal{P}(\mathcal{T} = v)$ denote the probability that token v is the correct (or most desirable) token in the "ground-truth" sense. Furthermore, let $P_\theta(v \mid x_{1:t-1})$ denote the language model's predicted token distribution conditioned on the given token sequence $x_{1:t-1}$, where θ are the language model parameters. Denoting the probability that the sampled token matches the ground-truth as $L[P]$, we have:

$$\begin{aligned} L[P] &:= \sum_{v \in \mathcal{V}} \mathcal{P}(\mathcal{S} = v \cap \mathcal{T} = v \mid x_{1:t-1}) \\ &= \sum_{v \in \mathcal{V}} \mathcal{P}(\mathcal{S} = v \mid x_{1:t-1}) \mathcal{P}(\mathcal{T} = v \mid x_{1:t-1}) \end{aligned} \quad (1)$$

since the sampling $\mathcal{S}$ and correctness $\mathcal{T}$ are independent events (no feedback involved). Notably, since we only have access to the predicted token distribution of the language model and no other external augmentation resources, we will take this as our best empirical estimate of the true token distribution, *i.e.* $\mathcal{P}(\mathcal{T} = v) = P_\theta(v \mid x_{1:t-1})$. Therefore, we have:

$$\begin{aligned} L[P] &= \sum_{v \in \mathcal{V}} \underbrace{\mathcal{P}(\mathcal{S} = v \mid x_{1:t-1})}_{=P_\theta(v \mid x_{1:t-1})} \underbrace{\mathcal{P}(\mathcal{T} = v \mid x_{1:t-1})}_{=P_\theta(v \mid x_{1:t-1})} \\ &= \sum_{v \in \mathcal{V}} P_\theta(v \mid x_{1:t-1})^2. \end{aligned} \quad (2)$$

We formalize the method as follows:

1. **Determine the threshold probability** $L[P_\theta]$ with equation 2.
2. **Construct the sampling set** $\mathcal{V}_{p\text{-less}}$ with tokens whose probabilities are at least $L[P_\theta]$:

$$\mathcal{V}_{p\text{-less}} = \{ v \in \mathcal{V} : P_\theta(v \mid x_{1:t-1}) \geq L[P_\theta] \}. \quad (3)$$

3. **Sample from** $\mathcal{V}_{p\text{-less}}$ the next token x_t according to the normalized token probabilities P'_θ :

$$P'_\theta(x_t \mid x_{1:t-1})|_{x_t=v} = \frac{P_\theta(v \mid x_{1:t-1})}{\sum_{v' \in \mathcal{V}_{p\text{-less}}} P_\theta(v' \mid x_{1:t-1})} \quad \text{for } v \in \mathcal{V}_{p\text{-less}}. \quad (4)$$

3.2 THE p -LESS_{NORM} SAMPLING METHOD

We further explore a variant of p -less which effectively relaxes the threshold by subtracting the likelihood of an incorrect random guess normalized to the number of correct outcomes. The resultant p -less_{norm}, denoted $\bar{L}[P_\theta]$, is preferable in use cases where diversity is favored over coherence. Formally, we have:

$$\bar{L}[P_\theta] := L[P_\theta] - \underbrace{\frac{1}{|\mathcal{V}| - 1}}_{\text{Normalization constant}} \times \underbrace{\sum_{u, v \in \mathcal{V}, u \neq v} P_\theta(u \mid x_{1:t-1}) P_\theta(v \mid x_{1:t-1})}_{\text{Probability of a randomly sampled and incorrect token}} \quad (5)$$

$$= \frac{|\mathcal{V}|}{|\mathcal{V}| - 1} L[P_\theta] - \frac{1}{|\mathcal{V}| - 1} \quad (6)$$

where $\frac{1}{|\mathcal{V}| - 1}$ gives the ratio of the possible number of correct to incorrect outcomes. The derivation of Eq. equation 6 from equation 5 and additional details of p -less_{norm} are provided in Appendix B.4.

3.3 CONNECTION TO RÉNYI ENTROPIES

Our p -less threshold can be re-interpreted in connection to established results in information theory, namely the family of Rényi entropies (Rényi, 1961). The Rényi entropy of order α ¹ is defined by:

$$H_\alpha(p) = \frac{1}{1-\alpha} \log \sum_i p_i^\alpha \quad (7)$$

In particular, the Rényi entropy of order 2 (aka the collision entropy) is given by

$$H_2(p) = -\log \sum_i p_i^2 = -\log L[P] \quad (8)$$

Since $\log$ is a monotonically increasing function, our p -less quantity $L[P]$ increases with decreasing collision entropy. Furthermore, we have

$$\begin{aligned} H_2(p) &= -\log L[P] \leq -\log \sum_i p_i \log p_i = H_1(p) \\ \implies L[P] &\geq \exp(-H_1(p)), \end{aligned} \quad (9)$$

which shows that $L[P]$ is also negatively correlated with the Shannon entropy.

The Rényi entropies of different orders quantify uncertainty with different sensitivities. In particular, the Rényi entropy of order 2 is sensitive to the concentration in the probability mass function and provides a suitable gauge of the global confidence in the model predictions.

Our p -less threshold corresponds to the exponential of the negative Rényi entropy of order 2. Intuitively, as Rényi entropy of order 2 increases, p -less decreases. By extension of the family of Rényi entropies, we also showed p -less to vary in the opposite direction as Shannon entropy (Shannon, 1948). Finally, p -less can be extended to a generalized k -order threshold within the formalism of Rényi entropy; see Appendix B.5 for details.

3.4 INTUITION OF p -LESS SAMPLING

We seek to answer the question "Given a probability distribution of tokens, what is a reliable subset of tokens we should sample from?" Rationally, we make use of the *full information in the distribution* to guide our decision, and formulate our method *principled in probability and statistics*. We define our threshold probability p -less, for filtering tokens into the sampling set, as the probability of a *randomly selected and correct* token (or most desirable token) in the "ground-truth" sense.

During inference, the token distribution that the large language model predicts is its degree of belief in the next-token ground-truth correctness, based on the ground-truth information it learned during training. Essentially, the token distribution encodes a notion of probabilistic correctness or desirability. This token distribution presents, via $L[P]$, the probability of sampling the ground truth, or equivalently the *probability that a random sampling is correct* (or most desirable), which we interpret as the minimum bar for tokens to qualify for admission into the sampling set. In other words, the tokens that we admit into the sampling set have to be *at least as confident as the random sampling that happens to be correct* (or most desirable) in the ground-truth sense.

¹For special values $\alpha \in \{0, 1, \infty\}$, the definition is extended via limits:

$$\begin{aligned} H_0(p) &= \log |\mathcal{V}| \\ H_1(p) &= -\sum_i p_i \log p_i \\ H_\infty(p) &= -\log \max p_i \end{aligned}$$

An alternative interpretation of $L[P]$ is that it serves as the unbiased estimator of the second moment of the distribution’s probability mass function, $M[P]$, scaled by the vocabulary size $|\mathcal{V}|$:

$$\begin{aligned} L[P] &:= \sum_{i=1}^{|\mathcal{V}|} P(x_i)^2 \\ &= |\mathcal{V}| \times \left\{ \frac{1}{|\mathcal{V}|} \sum_{i=1}^{|\mathcal{V}|} P(x_i)^2 \right\} \\ &= |\mathcal{V}| \times M[P] \end{aligned} \tag{10}$$

We regard p -less as an information-theoretic approach stronger than other methods, as it incorporates full information in the output token distribution for decoding, and demonstrates compatible interpretations using probability theory (see Sections 3.1 and 3.2), entropies (see Section 3.3) and statistical moments (see Section 3.4 and Appendix B.4). Specifically, p -less contrasts with other methods that do not consider the output token distribution (e.g. top- k , top- p , ϵ -sampling, min- p) or only considers it if conditions are met (e.g. η -sampling). p -less is also an empirical approach; it relies on the empirical output token distribution instead of making assumptions in the token distribution parameters or enforcing a target surprisal in the text generation process (e.g. mirostat), thereby eliminating further estimation errors.

3.5 ADVANTAGES OF p -LESS OVER EXISTING METHODS

p -less sampling combines several desirable properties of existing sampling approaches into a single method. First, the truncation threshold utilized in p -less sampling dynamically adapts to the entire token probability distribution at each time step. In contrast, existing sampling methods either use a fixed threshold which ignores the current token probability distribution (e.g. top- p , top- k , ϵ -sampling), set the threshold based on the probability of a single token in the current distribution (e.g. min- p), or only considers the token distribution if conditions are met (e.g. η -sampling). Second, p -less produces a bounded and valid truncation threshold which guarantees a non-empty candidate set for sampling, unlike other sampling methods where bounds are not guaranteed and edge cases are resolved with defaults, such as defaulting to the modal token (or top few tokens) if all tokens do not meet the threshold (e.g. ϵ -sampling, η -sampling, mirostat). Third, the truncation threshold of p -less sampling dynamically adjusts with temperature, unlike other methods (e.g. top- p , top- k , min- p , ϵ -sampling) whose hyperparameters are not meaningful when temperature approaches zero or infinity.

Thus, p -less uniquely possesses all three of the aforementioned desirable properties of a sampling approach, combining the benefits of existing sampling strategies into a single method. In addition, p -less is distinguished from prior approaches in that it is parameter-less. This eliminates the need to tune the sampling method’s hyperparameters, which are often sensitive to the generation task.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Our experiments were performed using Llama-2-7B (Chat) (Touvron et al., 2023), Mistral-7B (Instruct) (Jiang et al., 2023), and Llama3-70b (Instruct) (Dubey et al., 2024) on two types of tasks: *math and logical reasoning* across the GPQA (Rein et al., 2023), GSM8K (Cobbe et al., 2021), QASC (Khot et al., 2020) & CSQA (Talmor et al., 2019) datasets, and *instruction following creative writing* for the Writing Prompts (Fan et al., 2018a) dataset. We benchmarked our proposed sampling approaches against existing methods including Top- p (Holtzman et al.), Min- p (Nguyen et al., 2024), ϵ -sampling (Freitag et al., 2023), η -sampling (Hewitt et al., 2022) and Mirostat (Basu et al., 2021) for temperatures ranging from 0.5 to 2.0. We measured accuracy on the math and logical reasoning datasets and computed length-controlled win rate (Dubois et al., 2024) for Writing Prompts using an automated evaluation framework (Li et al., 2023), in addition to conducting a human evaluation. Additional experiment details are provided in Appendix C.1.

	Llama2-7b				Mistral-7b				Llama3-70b			
	CSQA	GPQA	GSM8K	QASC	CSQA	GPQA	GSM8K	QASC	CSQA	GPQA	GSM8K	QASC
ϵ	0.388	0.176	0.212	0.396	0.604	0.180	0.408	0.635	0.815	0.382	0.909	0.887
η	0.379	0.173	0.207	0.372	0.555	0.172	0.399	0.601	0.727	0.326	0.895	0.812
min- p	0.488	<u>0.243</u>	0.256	0.502	0.691	0.212	0.523	0.730	0.820	0.377	0.924	0.899
mirostat	0.410	0.212	0.201	0.419	0.635	0.216	0.392	0.684	0.776	0.366	0.879	0.879
top- p	0.410	0.172	0.210	0.393	0.580	0.172	0.438	0.604	0.713	0.320	0.870	0.778
p -less	0.503	0.242	0.267	<u>0.537</u>	0.697	0.239	<u>0.562</u>	<u>0.736</u>	0.819	<u>0.387</u>	0.932	<u>0.894</u>
p -less _{norm}	0.503	0.248	0.267	0.538	<u>0.692</u>	<u>0.222</u>	0.564	0.739	<u>0.819</u>	0.391	<u>0.930</u>	<u>0.894</u>

Table 1: AUC of Llama2-7b, Mistral-7b, and Llama3-70b across different sampling methods for math and logical reasoning datasets. The best AUC is in **bold** and the second best is underlined.

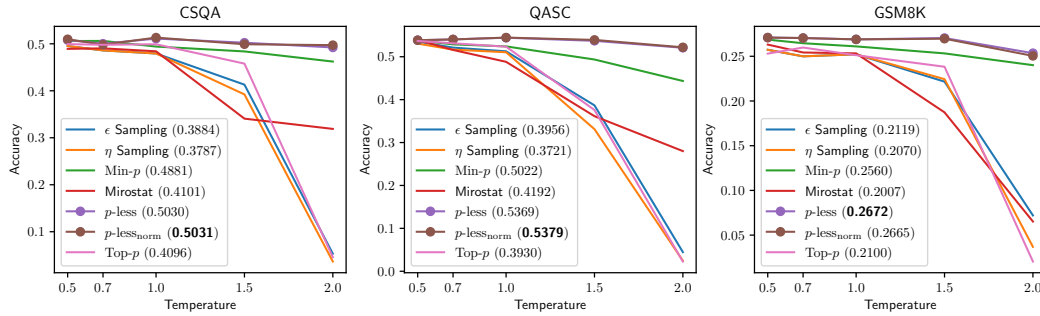


Figure 2: Accuracy vs. temperature curves of each method on CSQA, QASC, and GSM8k using Llama-2-7b. AUC values achieved by each method are provided in the legend (in parentheses) with the best AUC in **bold**.

4.2 MATH AND LOGICAL REASONING RESULTS

To perform a fair comparison between methods across temperatures, we computed the area under the accuracy-temperature curve for each method (normalized between 0.0 and 1.0), which we term AUC. Complete AUC results for the math & logical reasoning datasets are provided in Table 1. For Llama2-7b, the AUCs of p -less or p -less_{norm} outperform the other methods across all datasets. The results for Mistral-7b are consistent with those of Llama2-7b: the AUCs of both p -less and p -less_{norm} outperform all other methods across every dataset. For Llama3-70b, the AUCs of p -less and p -less_{norm} are either the highest or second highest within 0.005 of the highest. Across the four datasets on Llama2-7b and Mistral-7b, p -less and p -less_{norm} perform superior to the other methods at temperatures 1.0 and above, and are competitive at temperatures below 1.0 (see Figure 2, Table 5, Figure 9 and Figure 10). Figure 2 shows all sampling methods degrade at various rates with increasing temperature, while p -less and p -less_{norm} are robust to high temperatures and widen their performance gap against other methods. For Llama3-70b, p -less and p -less_{norm} perform superior to the other methods across all temperatures on GSM8K, on low and high temperatures for GPQA, and on high temperatures for CSQA and QASC; with the rest being competitive (see Table 5 and Figure 11).

In addition to the commonly-adopted default hyperparameter configurations we employed for the baseline methods (consistent with those employed or supported by prior work such as (Nguyen et al., 2024), (Zhou et al., 2024), (Zhu et al., 2024) and (Hewitt et al., 2022); see Appendix C.2 for the hyperparameter values), we conducted evaluations over a wide range of hyperparameter values for these baseline sampling methods (see Table 8 for full results), where we still find p -less and p -less_{norm} to generally have the highest AUC, and competitive to highest accuracies.

We also compared p -less at temperature 1.0 to greedy decoding and beam search using Mistral-7b, and provide a discussion on the versatility of p -less sampling in terms of producing strong results across a variety of tasks and datasets (see Appendices C.4, C.4.1 and Table 6). In low-entropy inference tasks which tend to favor greedy decoding (e.g. math and logical reasoning), p -less can achieve similar or better performance. Additionally, p -less performs significantly better than greedy decoding at higher-entropy tasks like creative writing, demonstrating that it is not simply an argmax-seeking procedure. Rather, p -less dynamically adapts to the entropy of the token distribution to produce high-quality outputs across a range of different settings without the need to change sampling strategies.

or hyperparameters depending upon the task. Furthermore, p -less can achieve higher diversity values (than greedy decoding) by increasing the temperature (see Appendix C.8 and Table 11).

We further ablated with the DeepSeek-R1-Distill-Qwen-7B (Guo et al., 2025) reasoning model, which consistently shows p -less and p -less_{norm} maintain strong performance across all temperature settings, including being significantly superior to the second best at temperature 2.0 (see Appendix C.5).

4.3 CREATIVE WRITING RESULTS

We provide results for the Writing Prompts dataset using the length-controlled win rate metric (Dubois et al., 2024). Specifically, we sample one generation per method and temperature for a subset of 100 prompts and use the response generated by default sampling (i.e. without truncation at temperature 1.0) as reference. Table 2 summarizes the results. All methods except p -less generally exhibit significant performance degradation as temperature increases. In contrast, the performance of p -less remains relatively stable and is superior to all other methods at temperatures > 1.0 .

This demonstrates how p -less excels in the domain of creative writing while avoiding the degradation of text quality exhibited by other methods at higher temperatures. In addition, we performed a human evaluation using the Llama2-7b generations for 100 sampled prompts, and obtained directional consistency between our human and automated evaluations, which provides further evidence of the effectiveness of p -less sampling for creative writing (see Appendix A).

Model	Temperature	ϵ -sampling	η -sampling	min- p	mirostat	top- p	p -less	p -less _{norm}
Llama-2-7b	1.0	62.18	58.76	57.48	56.94	62.07	55.08	58.74
	1.5	1.99	1.46	58.17	5.33	4.39	58.23	59.58
	2.0	0.00	0.00	48.94	26.88	0.00	65.64	59.29
Mistral-7b	1.0	60.90	59.82	66.49	62.26	65.68	68.90	67.49
	1.5	3.71	0.00	62.17	12.08	0.00	66.97	66.89
	2.0	0.00	0.00	54.11	40.33	0.00	60.32	61.99

Table 2: Length-controlled win rate for 100 sampled prompts from the Writing Prompts dataset.

5 ANALYSIS

5.1 EFFICIENCY OF p -LESS AND OTHER METHODS

To compare the inference-time efficiency of p -less to other sampling methods, we calculated the average sampling time per token over 200 Mistral-7b generations for GSM8K and GPQA. The results are summarized in Table 3. p -less achieves the fastest average sampling speed per token, with a 22% reduction in inference speed relative to min- p . The superior efficiency of p -less is statistically significant at the 5% level against the baseline sampling methods except η -sampling (see Appendix C.11 and Table 14). We attribute this greater efficiency to the fact that unlike other sampling approaches, p -less neither require sorting the token probability distribution to compute the truncation threshold, nor require determining the most confident token(s) for default inclusion into the sampling set in order to deal with the edge case of no tokens satisfying the truncation threshold. The other sampling approaches which we compare p -less to implement at least one of these two operations. Without the need to sort, p -less overall time complexity is reduced from $O(|\mathcal{V}| \log |\mathcal{V}|)$ to $O(|\mathcal{V}|)$; without the need to identify the most confident token(s), at least another $O(|\mathcal{V}|)$ operation is avoided for the case of the single most confident token. Additionally, we captured fine-grained CPU processing times and RAM usage during sampling for top- p , min- p and p -less; these results are consistent with Table 3, showing that p -less consumes the least CPU time and RAM (see Appendix C.11, Figures 16 and 17).

	ϵ -sampling	η -sampling	min- p	mirostat	top- p	p -less
Mean	0.02259	0.02210	0.02497	0.02278	0.02362	0.01942
Standard Deviation	0.01308	0.01277	0.01425	0.01339	0.00879	0.00899
Standard Error of Mean	0.0009	0.0014	0.0013	0.0013	0.0013	0.0009

Table 3: Average sampling time per token (in seconds) for p -less and other methods.

We also observed that p -less is often more efficient than other methods in terms of producing shorter generation lengths, despite achieving higher overall task-specific accuracy. Table 12 of Appendix C.9 provides the mean generation length produced by each sampling method across math and logical reasoning datasets, models and temperatures. In the case of Llama2-7b, p -less or p -less_{norm} produce the shortest generations on average across all temperatures for CSQA, QASC and GSM8K. These shorter generations do not sacrifice the model’s reasoning performance, as evidenced by the superior accuracy achieved by p -less and p -less_{norm} on these datasets while still being diverse (see Figures 2 and 3). These results provide empirical support to how our principled approach to truncation-based sampling improves both generation quality and inference-time efficiency. In Appendices C.9 and C.9.1, we provide a discussion on the generation lengths of top- p , min- p and p -less via the lenses of token probability distribution entropy and size of tokens admitted for sampling.

5.2 DIVERSITY ANALYSIS

We compute the n -gram repetition diversity metric proposed by Su et al. (2022) for QASC; higher values indicate greater diversity. Table 4 shows that at temperatures ≤ 1 , all methods produce similar diversity values between 0.62-0.64. At higher temperatures, p -less and p -less_{norm} exhibits similar diversity to min- p , but lower diversity than other sampling methods. However, greater diversity at these higher temperatures leads to lower answer accuracy. Figure 3 illustrates the relationship between generation diversity and answer accuracy for Llama2-7b using sampling methods and temperature settings which achieved an overall mean accuracy > 0.5 on QASC. This plot shows that p -less and p -less_{norm} produce higher accuracy at a given level of generation diversity than other sampling methods, exhibiting a pareto dominance along the diversity-accuracy frontier. We provide additional diversity results for all three models on the four math and logical reasoning datasets in Table 10 of Appendix C.8. These results show that as temperature increases, p -less exhibits a reasonable increase in diversity while other methods experience diversity spikes which compromise their task-specific reasoning capability (see Table 5). We further ablated p -less with temperature values > 2.0 , and show that p -less allows for significant increase in diversity based on temperature; it exhibits a similar increasing trend of diversity with temperature as other methods such as min- p , differing primarily in the magnitude of diversity increase with each temperature step (see Appendix C.8 and Table 11).

Temperature:	0.5	0.7	1.0	1.5	2.0
ϵ Sampling	0.63	0.63	0.63	0.75	0.98
η Sampling	0.62	0.63	0.63	0.79	1.00
Min- p	0.63	0.62	0.62	0.62	0.64
Mirostat	0.62	0.63	0.63	0.79	0.76
Top- p	0.63	0.63	0.62	0.73	1.00
p -less	0.63	0.64	0.63	0.63	0.64
p -less _{norm}	0.63	0.64	0.63	0.63	0.64

Table 4: QASC diversity by method & temperature

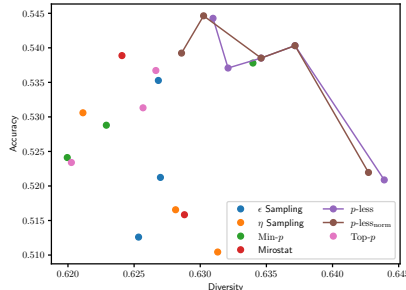


Figure 3: QASC accuracy vs. diversity

5.3 QUALITATIVE ANALYSIS

We found that where the other methods fail on specific math and logical reasoning examples at low temperatures, p -less and p -less_{norm} generate correct reasoning and the correct final answer. When temperatures are high enough for the methods to fail, p -less and p -less_{norm} still generate correct reasoning, including unprompted self-verification and self-correction on initial wrong answers. At high temperatures, the generations of other methods veer off-track or exhibit hallucination and degeneracy, whereas generations from p -less and p -less_{norm} remain factual and relevant. Illustrative examples are discussed in Appendix C.12.2. We also provide creative writing examples where p -less is preferred to default sampling by human evaluators and the automated evaluation framework in Appendix C.12.3. Finally, we discuss two typical failure patterns of p -less in Appendix C.13.

5.4 CASE STUDY: ROBUSTNESS UNDER HIGH ENTROPY

GSM8K example at temperature 2.0: We examine responses to a question which involves interpreting the order of events, where the response produced by min- p committed a critical misinterpretation error. We supplement the responses with a plot of the entropy and number of admitted tokens at each generation step. Notably, even though the average entropy and fluctuation in entropy is significantly smaller for min- p than p -less, the number of admitted tokens after the truncation is higher. This contrast illustrates how p -less enforces a form of entropy-aware regularization, mitigating token overcommitment in ambiguous regions and preserving semantic fidelity. We delve into further quantitative analysis of the entropy and token count distributions in Appendix C.10.

Question: "Brian's friend Bobby has 5 fewer than 3 times as many video games as Brian does. If Brian has 20 video games but lost 5 right before the comparison was made, how many does Bobby have?"

Min- p : *Let's think step by step. First, we need to find out how many video games Brian had originally. If he has 20 video games now and lost 5, then he must have had $20 + 5 = \langle 20+5=25 \rangle 25$ video games originally. Bobby has 3 times as many video games as Brian has, minus 5. So Bobby has $3 * 25 - 5 = 75 - 5 = \langle 75-5=70 \rangle 70$ video games. The answer is 70.*

The flawed reasoning step in red above coincides with a region of spikes in the number of admitted tokens (circled region) in Figure 4a. Separately, at the mid-point of the generation, there is an unexpected large spike in token admission despite a corresponding ordinary entropy.

p -less: *Let's think step by step. Brian had 20 video games, but lost 5, so he has $20 - 5 = 15$ video games now. Bobby has 3 times as many video games as Brian, so Bobby has $3 * 15 = 45$ video games. But Bobby has 5 fewer than 3 times as many video games as Brian, so Bobby has $45 - 5 = 40$ video games. The answer is 40.*

As seen in Figure 4b, even while the entropy is extraordinarily high, the number of admitted tokens remains well-regulated, showing p -less's selectivity in (desirable) tokens and effectiveness in subduing verbosity.

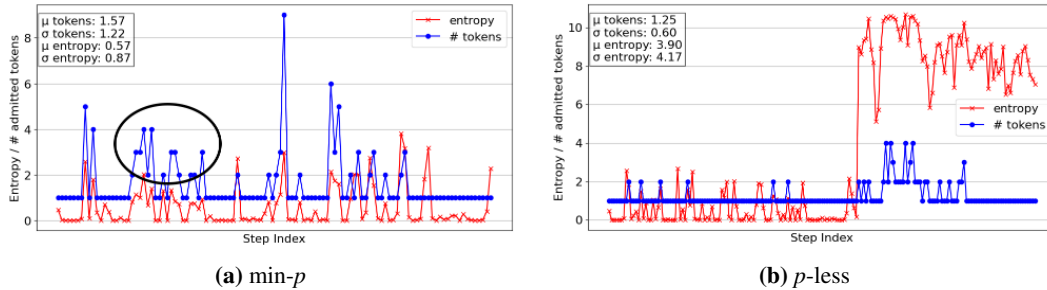


Figure 4: Step-wise entropy and number of admitted tokens for a GSM8K question answered with Llama3-70b.

6 CONCLUSION

We presented p -less sampling: a hyperparameter-free truncation methodology for sampling-based decoding. p -less combines several desirable properties of existing sampling methods into a single approach while eliminating the need to tune sampling hyperparameters required by other methods. Our experimental results across three LLMs and five datasets spanning math, logical reasoning, and creative writing tasks demonstrated how p -less consistently achieves strong performance across a range of different temperature values. In contrast, other evaluated methods exhibit significant declines in performance as temperature increases. We further showed how p -less achieves greater inference-time efficiency than other methods through faster token sampling speeds and shorter generation lengths. Our work highlights how grounding LLM decoding in information theory results in a principled sampling approach which is both intuitive and empirically effective.

7 REPRODUCIBILITY STATEMENT

We will make our source code publicly available upon publication in order to facilitate future efforts to reproduce our main experimental results. In addition, we have provided complete details of models, datasets, and evaluation metrics in Appendix C.1. We provide details on hyperparameters utilized in our experiments in Appendix C.1 and Appendix C.2. From Appendix C.3 to Appendix C.11, we detailed our methods on analyses. In Appendix C.12 and Appendix C.13, we provide details of prompts used in our experiments and provide multiple generation examples. To the best of our knowledge, the documentation in this manuscript contains all details necessary to fully reproduce our results.

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A HUMAN EVALUATION

As the four math and logical reasoning datasets contain annotated labels and will therefore not benefit significantly from a human evaluation, we focused our human annotation study on the creative writing task. Specifically, three authors of this work and three non-author annotators evaluated Llama2-7b generations produced by p -less and default sampling for 100 sampled prompts from the Writing Prompts dataset, producing a total of 4 labels for each story pair. The p -less samples were generated with a temperature of 2.0 and were evaluated in the same pairwise comparison setting as our length-controlled win rate evaluations. 23.7% of story pairs received unanimous agreement among the annotators and 26.9% received a tie; for the remaining stories, we use the majority vote to obtain a label. Overall we found that p -less won the majority vote 58.8% of the time, with the win rate further increasing to 72.7% for the 23.7% of stories which had unanimous agreement. The win rates for annotations produced by authors were 57.6%, 54.3%, 57.1% while the win rate for the non-author annotations was 54.9% (these values are slightly lower than the majority vote win rate due to the presence of ties), indicating that the labels produced by authors and non-author annotators are distributionally similar. The directional consistency of our human and automated evaluations (Table 2) provides further evidence of the effectiveness of p -less sampling for creative writing.

B ADDITIONAL PROPERTIES OF p -LESS AND p -LESS_{NORM} SAMPLING

B.1 FRIEDMAN’S INDEX (PROBABILITY) OF COINCIDENCE

In cryptography, the Friedman’s Index (Probability) of Coincidence (Friedman, 1922), IC , for an infinitely long encryption can be approximated with the p -less quantity.

$$\begin{aligned}
 IC &= \frac{\sum_i n_i(n_i - 1)}{N(N - 1)} \\
 \lim_{n \rightarrow \infty} IC &= \lim_{n \rightarrow \infty} \frac{\sum_i n_i(n_i - 1)}{N(N - 1)} \\
 &= \lim_{n \rightarrow \infty} \sum_i \left(\frac{n_i}{N} \right) \left(\frac{n_i - 1}{N - 1} \right) \\
 &= \lim_{n \rightarrow \infty} \sum_i (p_i) \left(\frac{n_i - 1}{N - 1} \right) \\
 &= \sum_i (p_i) (p_i) \\
 &= \sum_i p_i^2
 \end{aligned} \tag{11}$$

B.2 UNBIASED ESTIMATOR OF THE SECOND MOMENT OF THE PROBABILITY MASS FUNCTION

p -less is also the unbiased estimator of the second moment of the probability mass function of the distribution, $M[P]$, multiplied by the vocabulary size $|\mathcal{V}|$:

$$\begin{aligned}
 L[P] &:= \sum_{v \in \mathcal{V}} P(v \mid x_{1:t-1})^2 \\
 &= |\mathcal{V}| \times \left\{ \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} P(v \mid x_{1:t-1})^2 \right\} \\
 &= |\mathcal{V}| \times M[P] \\
 &\propto M[P]
 \end{aligned} \tag{12}$$

This demonstrates that the p -less quantity $L[P]$ is directly proportional to the unbiased estimator of the second moment of the probability mass function of the token distribution $M[P]$. Notably, as the distribution becomes more uniform (having more entropy), its second moment decreases, and the p -less quantity decreases. With a smaller p -less quantity, the method intuitively admits more tokens.

Our principled approach in deriving and proposing the use of p -less is further supported by these connections. We invite the community to explore further methods grounded in information theory.

B.3 DESIRABLE PROPERTIES OF p -LESS SAMPLING

We compute a threshold probability for filtering tokens from the large language model’s output token distribution for sampling, grounded on the *likelihood of a correct random guess given the distribution*, as a principled way to decide what to sample.